

Lesson 6 Combinatorics,

Principles of Counting

(without counting!)

Probability calculations often involve counting, especially when

we have equally likely outcomes and finite sets. In such

situations, the probability of

an event is just the "size"

(cardinality) of that event

relative to the size

(cardinality) of the sample

space i.e.:

$$P(A) = \frac{|A|}{|\Omega|} \quad \forall A \subseteq \Omega$$

In this lecture, we review the

concepts of combinatorics, for

measuring the "size" of countable events.

The Multiplication Principle

Consider a two-stage experiment.

the first stage can have

m results $a_1, a_2, \dots, a_m$

and the second stage can

have n results $b_1, b_2, \dots, b_n$.

The number of possible ways

to perform the first and the

second stage sequentially is

the number of possible (a_i, b_j)

pairs, which is

$$\boxed{mn}$$

More formally:

A, B finite

If $\Omega = A \times B$, then

$$|\Omega| = |A \times B| = |A| |B|$$

More generally

If a process has r stages

and stage i has n_i

possible results, the number
of all possible results is

$$n_1 n_2 \cdots n_r = \prod_{i=1}^r n_i$$

In more formal terms, if

$\Omega = A_1 \times A_2 \times \dots \times A_r$, then

$$|\Omega| = |A_1 \times A_2 \times \dots \times A_r| = |A_1| |A_2| |A_3| \dots |A_r|$$

$$\text{--- } |A_r| = \prod_{i=1}^r |A_i|$$

$$\text{If } \forall i, A_i = A, \quad |\Omega| = |A^k| = |A|^k$$

Example: Telephone numbers are 7 digits in Springfield.

The first digit cannot be 1 or 0. How many different telephone numbers can we have?

Solution:

$$8 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 8 \times 10^6$$

Example: What is the number of subsets of a set with n elements?

$$\left\{ \begin{array}{ccccccc} \emptyset & \circ & \circ & \circ & \circ & \text{---} & \circ \\ A & 1 & 1 & 1 & \text{---} & 1 & \\ & \underbrace{2 \times 2 \times 2} & & & \underbrace{\times 2} & = 2^n \end{array} \right.$$

choice

$$|2^{\Omega}| = 2^{|\Omega|}$$

The multiplication principle helps us understand ordered sampling with replacement.

Example: Assume that we have n objects, and we would like to

sample k objects from them with replacement. Each time, we draw an object, make a note of it, put the object back and select the next object.

The number of different sequences of k objects that can be formed is:

$$\underbrace{n \times n \times \dots \times n}_{k \text{ times}} = n^k$$

Permutations: ~~The~~ When k objects are selected from a plurality of n objects, and order matters, k -permutations are constructed.

In other words, k -permutations
 are subsequences of length k , selected
 from sequences of length n

k places

$$n \times n-1 \times n-2 \times \dots \times n-k+1 = \frac{n!}{(n-k)!}$$

$$n \times (n-1) \times \dots \times (n-k+1) \times (n-k)! = n!$$

In other words, permutations can

be considered as results of ordered
sampling without replacement

$$\Omega = A \times (A - \{w_1\}) \times (A - \{w_1, w_2\}) \times \dots \times (A - \{w_1, \dots, w_{k-1}\})$$

$$|\Omega| = \frac{n!}{(n-k)!}$$

Order matters $(1, 2, 3) \neq (3, 2, 1)$

$\neq (2, 3, 1)$

Example : Assume that

we would like to enumerate
all possible ^{binary} sequences ~~sub~~ that

can be created using (0,0,1,1).

$$m = 4$$

$$k = 3$$

$$\omega_1 =$$

$$\omega_2 =$$

$$\omega_3 =$$

$$\omega_4 =$$

Observe that $\square$ is a

rearrangement of $\square$ and

$\square$ is a rearrangement of $\square$

Combinations: when^{we} select~~the~~

k objects from n objects, and

order does not matter for us,

we are considering combinations.

"Order does not matter." This

means that all re-arrangements

of a sequence of length

k should be considered

the same sequence.

Question: How many

rearrangements of a sequence
of length k exist?

k places $\overbrace{k \quad k-1 \quad k-2 \quad \dots \quad 1} = k!$
 k objects

$$\frac{k!}{(k-k)!} = \frac{k!}{0!} = k!$$

number of permutations

Therefore, the number of
combinations of length k
from n objects is

$$\binom{n}{k} = \frac{\# \text{ permutations}}{k!} = \frac{n!}{k! (n-k)!}$$

Combinations are useful

in understanding "unordered

sampling without replacement"

$\binom{n}{k}$ is also called the

binomial coefficient, because it

appears in the binomial theorem:

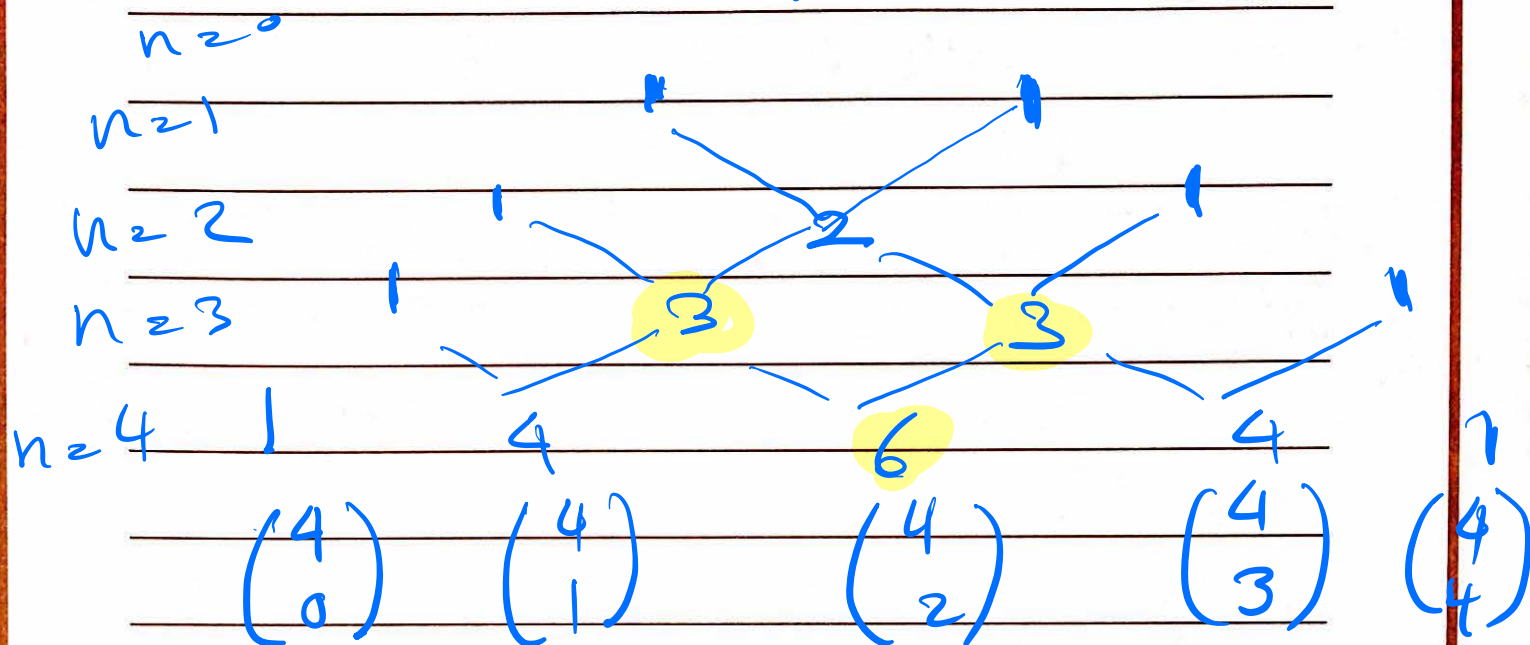
$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2$$

$$+ \dots + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n = \sum_{k=0}^n \binom{n}{k}a^{n-k}b^k$$

The coefficients $\binom{n}{k}$ can

be read from the so-called

Khayyam-Pascal Triangle



~~Pass~~ The triangle works based on the following formula:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

The binomial theorem can be proved by induction.

Left as an exercise.

Example: We have 5 novels,
3 Sci-fi books, 4 magazines,
and 8 college textbooks.

In how many different ways
can they be arranged so that

books of the same genre
are placed together?

novels 5!

sci-fi 3!

mag 4!

textbooks 8!

$$5! \times 3! \times 4! \times 8! \times 4!$$

Partitions A combination is

a partition of n objects $\binom{n}{k, n-k}$
into k and $n-k$ objects.

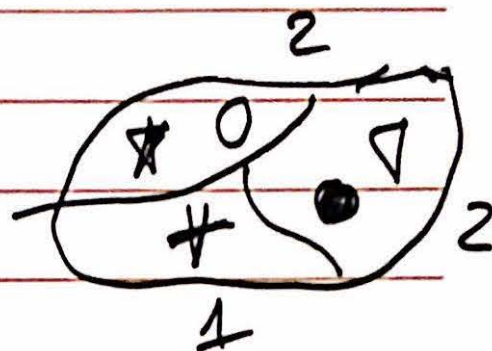
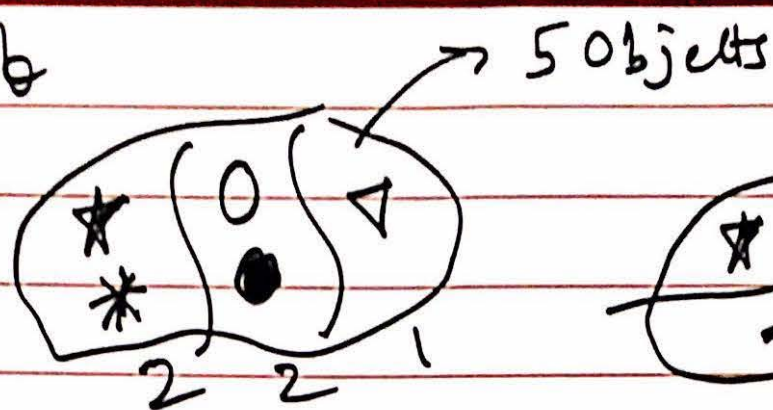
We can generalize this discussion
to dividing

n objects into r partitions,
given that the i^{th} partition
has n_i elements.

Obviously

$$n_1 + n_2 + \dots + n_r = n$$

Example



Number of such partitions is

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!} \quad \text{induction on } r$$

We ^{also} use the notation:

$$\binom{n}{n_1, n_2, \dots, n_r}$$

Interestingly,

$$\binom{n}{n_1 \ n_2 \ \dots \ n_r}$$

is called a "multinomial coefficient," because it appears in the multinomial theorem

$$(a_1 + a_2)^n = \sum_k \binom{n}{k, n-k} a_1^k a_2^{n-k}$$

$$(a_1 + a_2 + \dots + a_m)^n = \sum_{k_1, k_2, \dots, k_m} \binom{n}{k_1, \dots, k_m}$$

$$k_1 + k_2 + \dots + k_m = n$$
$$a_1^{k_1} a_2^{k_2} \dots a_m^{k_m}$$

Try $(a_1 + a_2 + a_3)^4$

Example (Anagrams)

How many different words
(letter sequences) can be
obtained by rearranging
the letters in the word

TATTOO?

T O A O T T

